RG RESTAURANT

FOOD DELIVERY OPERATIONS ANALYTICS

Project Report

Figure 1. RG Restaurant Power BI landing page

Using SQL • Python • Power BI

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1. Executive Summary

The Food Delivery Operations Analytics project was developed for RG Restaurant to understand sales performance, customer behavior, menu performance, delivery-partner efficiency, and delivery delays. The solution combines SQL for structured business analysis, Python for data cleaning and exploratory analysis, and Power BI for interactive visualization and management reporting.

The project works with five related datasets: Orders, Menu Items, Customers, Delivery Partners, and Reviews. The source workbook contains 1,200 order records, 25 menu items, 300 customers, 25 delivery partners, and 598 reviews. The project flow follows the required sequence of SQL database analysis, Python cleaning and EDA, derived KPIs, Power BI modeling, dashboard development, and business insights.

2. Problem Statement

Restaurant delivery operations generate data across orders, customers, menu items, delivery partners and reviews. Without a consolidated analytical view, it is difficult to identify where sales are strongest, which items drive revenue, which customer segments contribute most, and what operational factors are associated with delivery delays.

The project therefore focuses on measuring delivery-time performance, identifying delay drivers, evaluating menu-category and delivery-partner performance, and analyzing peak-hour order and customer trends.

3. Project Objectives

- Measure overall delivery performance and delivery delay rate.
- Identify delivery areas and operating conditions associated with delays.
- Analyze sales, order volume, average order value and category contribution.
- Identify top-performing menu items and evaluate item performance using rating, volume and delay rate.
- Evaluate delivery partners using order volume, delivery performance and customer rating.
- Understand customer segments, gender distribution, payment modes and area-wise revenue.
- Analyze hourly, weekly and monthly order patterns to support operational planning.
- Build an interactive Power BI dashboard with slicers, KPIs and management insights.

4. Dataset Description

The project uses five structured tables. The dataset schema and project task sheet define Orders, Menu Items, Customers, Delivery Partners and Reviews as the core sources.

Table	Rows	Main Purpose
Orders	1,200	Order, sales, delivery, payment and rating details
Menu Items	25	Item category, pricing, cost, margin and bestseller attributes
Customers	300	Customer demographics, area and segment
Delivery Partners	25	Partner, vehicle type, experience and rating
Reviews	598	Order/customer review, rating, date and sentiment

Important Orders fields include order date/time, peak-hour flag, delivery area, item, quantity, order value, distance, promised delivery time, actual delivery time, delay, order status, delivery partner and customer rating.

5. Tools & Technologies

Technology	Usage
SQL / MySQL	Database creation, data validation, joins, aggregation, ranking and business queries
Python	Data loading, cleaning, feature engineering and exploratory data

	analysis
Pandas / NumPy	Data manipulation and derived metrics
Matplotlib / Seaborn	EDA charts and analytical visualizations
Power BI	Data modeling, DAX KPIs, slicers and interactive dashboards
Excel	Original multi-sheet dataset source
GitHub	Project version control and portfolio submission

6. Project Methodology

The project follows an end-to-end analytics workflow:

- Source Excel workbook → SQL database
- SQL validation and business analysis
- Python data cleaning and exploratory analysis
- Derived delivery and performance metrics
- Power BI data model and DAX KPIs
- Interactive dashboard development
- Management insights and business recommendations

Figure 2. Executive Overview dashboard

7. Data Cleaning & Preparation

The Python notebook begins by loading all five Excel sheets with pandas and checking shape, schema, descriptive statistics, missing values and duplicates. The observed row counts are 1,200 Orders, 25 Menu Items, 300 Customers, 25 Delivery Partners and 598 Reviews.

Missing-value analysis identified missing delivery-related fields in Orders. The notebook shows 42 missing actual delivery times, 88 missing delay values, 88 missing Is Delayed values, 42 missing delivery partner IDs and 88 missing customer ratings.

These missing values are associated with records that do not have complete delivery outcomes, so delivery metrics are analyzed using delivered orders.

No exact duplicate rows were detected in the five tables, and the primary identifier checks in the notebook returned zero duplicates for order_id, item_id, customer_id, partner_id and review_id.

Text standardization was performed using trimming and whitespace normalization across delivery area, category, payment mode, order status, peak-hour flag, customer segment, vehicle type and review sentiment fields. Date fields were converted to datetime, order time was converted to time, and order hour was extracted.

The notebook derives delay_mins from actual delivery time versus promised delivery time for delivered orders, creates an Is Delayed flag, calculates a Delivery Speed Score, and builds item-level and partner-level performance summaries.

8. SQL Analysis

A MySQL database named food_analytics was created and analytical queries were written against cleaned_orders. The SQL work includes KPI calculation, revenue analysis, top-item analysis and delivery-area delay analysis.

Analysis	SQL Logic / Purpose
Total Orders	COUNT(*) on cleaned_orders
Total Revenue	SUM(total_amount_inr) for Delivered orders
Average Delivery Time	AVG(actual_delivery_mins) for Delivered orders
Delay Rate	Delayed delivered orders ÷ delivered orders × 100
Top 10 Items	Group by item_name and order revenue descending
Delay Rate by Area	Delivered orders grouped by delivery_area and ranked by delay rate

The SQL project specification also calls for joins across Orders, Menu Items, Customers and Delivery Partners, category-level KPIs, peak/non-peak analysis, partner ranking, customer segmentation and a consolidated Power BI view.

Figure 3. Sales & Customer Analysis dashboard

9. Python EDA & Analytics

Python exploratory analysis examined delivery-time, distance, order-value and customer-rating distributions, followed by area, item, category and partner comparisons. The notebook creates the required analytical charts for delay rate by area, top items by revenue, hourly order trend, distance versus delay and delivery-partner performance.

Python Analysis	Observed Result
Average order value	₹407.15
Average distance	6.41 km
Average promised delivery	44.29 minutes
Average actual delivery	47.77 minutes in notebook; 47.81 minutes on dashboard
Average delay	6.66 minutes among records with delay values
Average customer rating	3.85 in notebook
Top revenue item	Mutton Rogan Josh — ₹48,320

The item performance logic classifies items as High, Medium or Low using rating, order volume and delay rate thresholds based on median item performance. In the notebook output, Mutton Rogan Josh is classified as Medium performance with 55 orders, ₹48,320 revenue, 4.18 average rating and 61.82% delay rate.

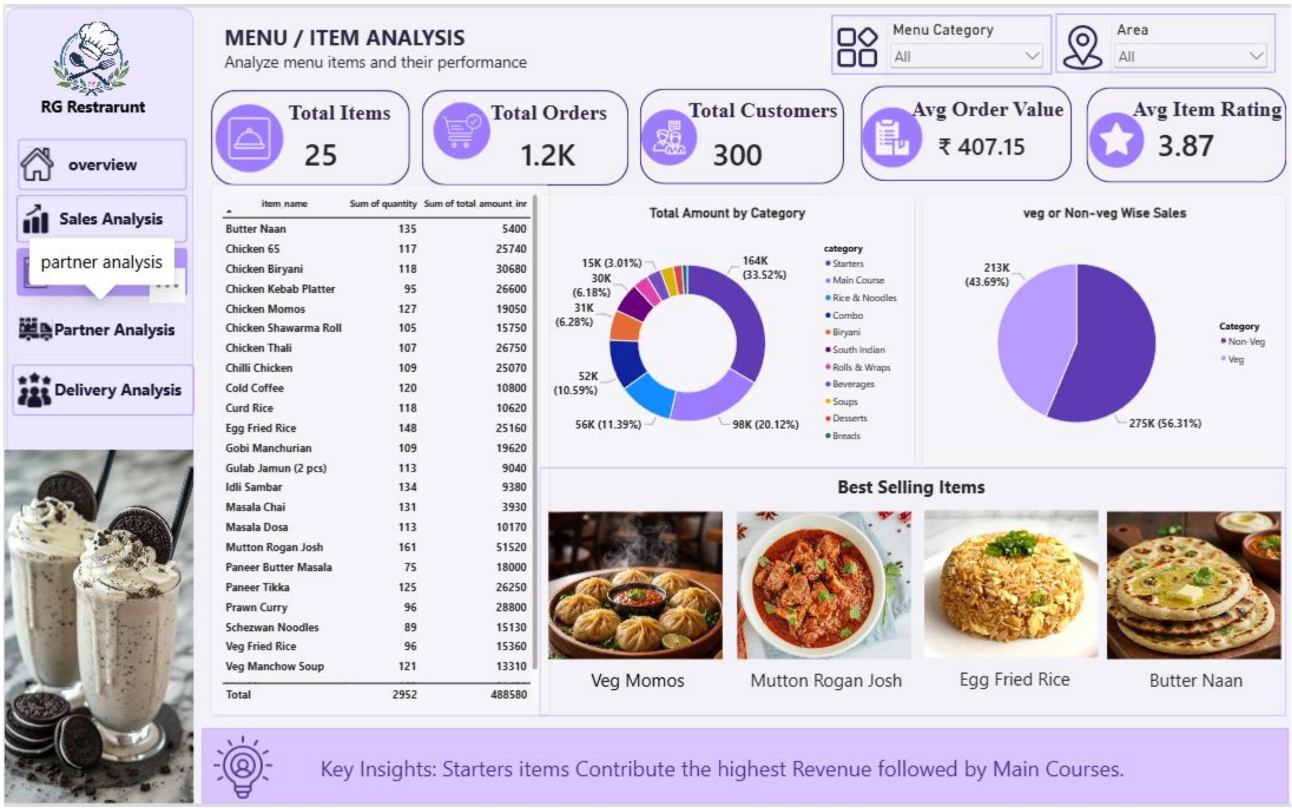

Figure 4. Menu / Item Analysis dashboard

10. Power BI Dashboard Design

The Power BI solution uses a consistent lavender visual theme with KPI cards, charts, slicers, navigation buttons and an insight strip. The dashboard is designed for management-level exploration rather than only static reporting.

Page	Purpose	Main Visuals / KPIs
Welcome	Landing and navigation	RG Restaurant branding, Explore Dashboard button
Executive Overview	Overall performance	Sales, orders, customers, AOV, rating, monthly trend, customer segment revenue, area revenue, top item
Sales & Customer Analysis	Sales and customer behavior	Gender, category revenue, payment mode, area rating and order analysis
Menu / Item Analysis	Menu performance	Item table, category revenue, veg/non-veg

		sales, best-selling items
Delivery Partner Analysis	Partner performance	Partner orders, delivery time, vehicle revenue, partner ratings
Delivery Analysis	Delivery operations	Hourly/weekly orders, customer segments, area/peak-hour views and delivery KPIs

Slicers visible across the dashboard include date/join-date range, area, payment mode, partner name, customer segment, menu category and vehicle type, depending on the selected page.

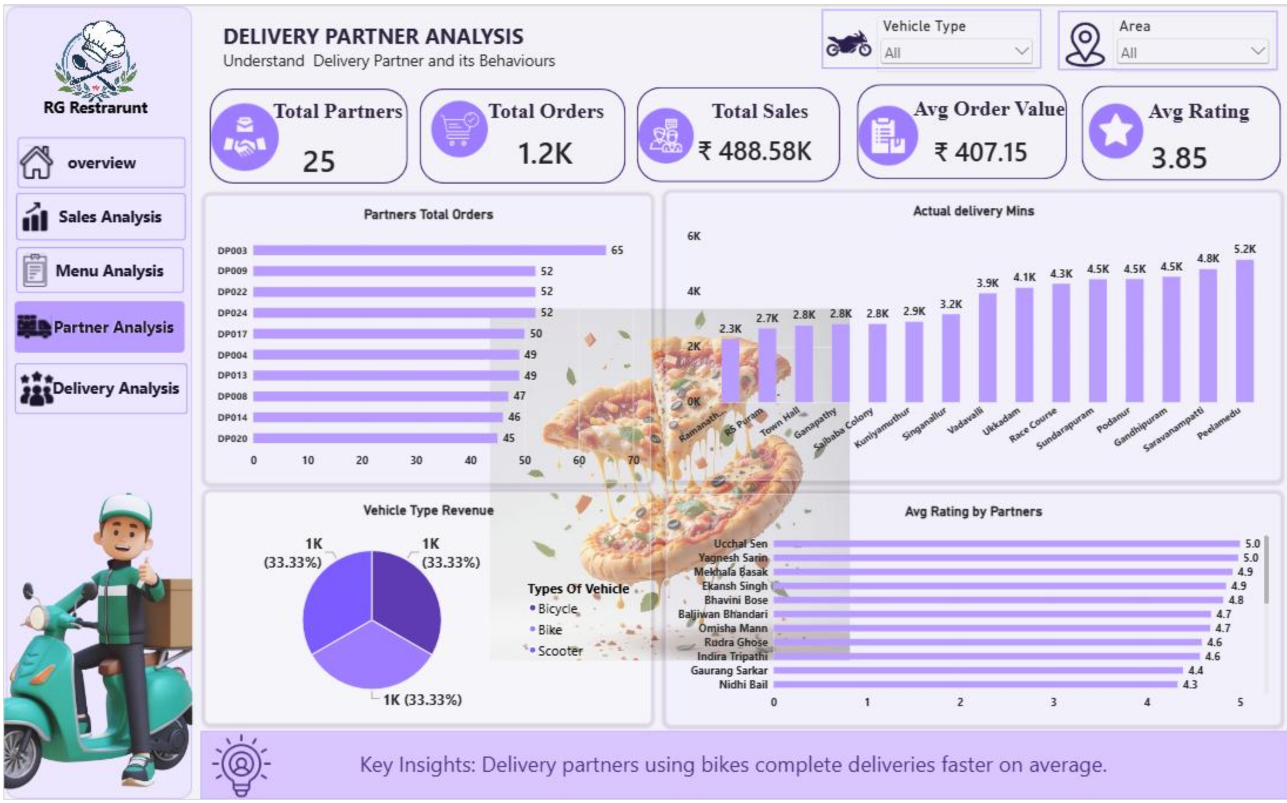

Figure 5. Delivery Partner Analysis dashboard

Figure 6. Delivery Analysis dashboard

11. Dashboard Findings

11.1 Overall Performance

The Executive Overview reports ₹488.58K in total sales from 1.2K orders and 300 customers, with an average order value of ₹407.15 and an average customer rating of 3.85. Monthly order activity varies across the displayed January–September period, indicating opportunities to align staffing and promotions with demand changes.

11.2 Sales & Customer Findings

The dashboard shows Regular customers contributing the largest share of revenue at approximately ₹187K (41.21%), followed by Loyal customers at ₹103K (22.79%), New customers at ₹120K (26.42%) and VIP customers at ₹43K (9.58%).

Digital payment methods dominate order activity: UPI records 311 orders, Credit/Debit Card 308, Wallet 300, and Cash on Delivery 281. Female customers account for 164 customers (54.67%) and male customers account for 136 (45.33%).

11.3 Menu & Item Findings

Starters are the largest revenue category in the dashboard at about ₹149K (32.78%), followed by Main Course at ₹92K (20.26%). The dashboard highlights Mutton Rogan Josh as the top-selling item, and the Python analysis records ₹48,320 revenue and 55 orders for this item.

The menu page also highlights best-selling items visually, including Veg Momos, Mutton Rogan Josh, Egg Fried Rice and Butter Naan.

11.4 Delivery Partner Findings

The dashboard covers 25 delivery partners and reports ₹488.58K total sales, 1.2K total orders, ₹407.15 average order value and a 3.85 average rating. Partner order volumes vary, with DP003 shown at the top of the visible partner-order ranking with 65 orders.

The dashboard's key insight states that delivery partners using bikes complete deliveries faster on average. This supports vehicle-based operational planning, especially for higher-demand delivery periods.

11.5 Delivery Findings

Average delivery time is approximately 47.81 minutes. The weekly chart shows Thursday with the highest displayed order volume at 185 orders, followed by Monday at 182 and Wednesday at 177; Sunday is the lowest displayed day at 144.

The dashboard's key insight identifies traffic and distance as the main reasons for delivery delays. This aligns with the Python analysis, which explicitly studies the relationship between distance and delivery delay.

The hourly order trend shows demand across the operating hours, supporting peak-hour staffing and delivery-partner allocation.

12. Business Recommendations

- Use traffic-aware route planning and prioritize shorter delivery routes during high-demand periods because the dashboard identifies traffic and distance as key delay drivers.
- Increase the availability of bike-based delivery partners during peak periods where faster delivery performance is required.
- Use area-level delay monitoring to identify consistently high-delay locations and adjust partner allocation or promised delivery times accordingly.
- Promote high-revenue menu categories, especially Starters and Main Course items, while maintaining visibility for proven best-selling items.
- Use Mutton Rogan Josh as a promotional anchor because it is the top revenue item in the analysis; monitor its delay rate before scaling promotions.
- Strengthen retention campaigns for Regular and Loyal customers because these segments contribute the largest combined revenue share.
- Encourage digital payments through offers or loyalty benefits, since UPI, cards and wallets together account for most displayed orders.
- Use hourly and weekly order trends to schedule kitchen capacity and delivery-partner availability more efficiently.

13. Challenges & Solutions

Challenge	Solution Applied
Missing delivery fields	Analyzed delivery KPIs on completed/delivered orders and derived delay metrics only where delivery outcomes exist.
Inconsistent text formatting risk	Applied strip and whitespace normalization to categorical/text fields.
Multiple business dimensions	Used SQL joins and Power BI relationships to connect orders with customers, menu and delivery partners.
Operational delay analysis	Created delay minutes, delayed flag, speed score and partner/item performance summaries.
Management readability	Used KPI cards, slicers, navigation, consistent theme and insight callouts in Power BI.

14. Conclusion

The RG Restaurant Food Delivery Operations Analytics project converts raw operational data into a structured decision-support solution. SQL provides the analytical foundation, Python improves data quality and uncovers patterns, and Power BI presents the results through an interactive management dashboard.

The analysis highlights strong sales contribution from Regular customers and Starters, strong performance from selected menu items, dominant digital payment usage, variation in partner performance, and delivery delays associated with traffic and distance. These findings provide practical directions for route planning, partner allocation, menu promotion and customer retention.

15. Future Scope

- Add real-time GPS and traffic data for more accurate delay prediction.
- Build a predictive model for expected delivery time and late-order probability.
- Introduce customer lifetime value and churn prediction.
- Track profit rather than revenue alone by combining item margin with order volume.
- Add automated alerts for high-delay areas, low-rated partners and stock risks.
- Publish scheduled management reports through a BI reporting workflow.

16. References / Project Artifacts

Project artifacts used for this report include the Food Delivery Operations Analytics task specification, the Python analysis notebook, the SQL analysis script, and the completed Power BI dashboard screenshots.

Artifact	Purpose
food delivery operation analytics.ipynb	Python data loading, cleaning, EDA and derived analytics
food_operation_analytics.sql	Database setup and analytical SQL queries
Food _delivery_operation_analytics.pbix	Power BI data model, DAX and interactive dashboard
Dashboard screenshots	Visual evidence of the completed report pages

Project title: Food Delivery Operations Analytics Using SQL, Python & Power BI

Domain: Food-Tech / Single-Restaurant Delivery Operations Analytics